MILESTONE 6 · FINAL SYSTEM, INTEGRATION & DEPLOYMENT

AI-Powered Driver Wellness & Safety Monitoring System

Five trained models unified behind one standardized contract, fused into a live 0–100 wellness score, and deployed as a hosted real-time driver-monitoring application.

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BSDA4001 · DS/AI Lab Course · Final Project Presentation

THE FULL ARC

Six Milestones, One System

From a revised scope to a fully fused, deployed real-time wellness engine — each milestone built directly on the last.

M1

Scope & Framing

Reframed beyond drowsiness → full driver-wellness; multi-branch pipeline proposed.

M2

Dataset Prep

Five datasets cleaned & split leakage-free — subject / sequence-wise.

M3

Architecture

One model per module selected, justified & fully specified; fusion designed.

M4

Train & Tune

Five checkpoints trained; hyperparameters swept per module.

M5

Held-Out Eval

Benchmarked, error-diagnosed & efficiency-measured on unseen data.

M6

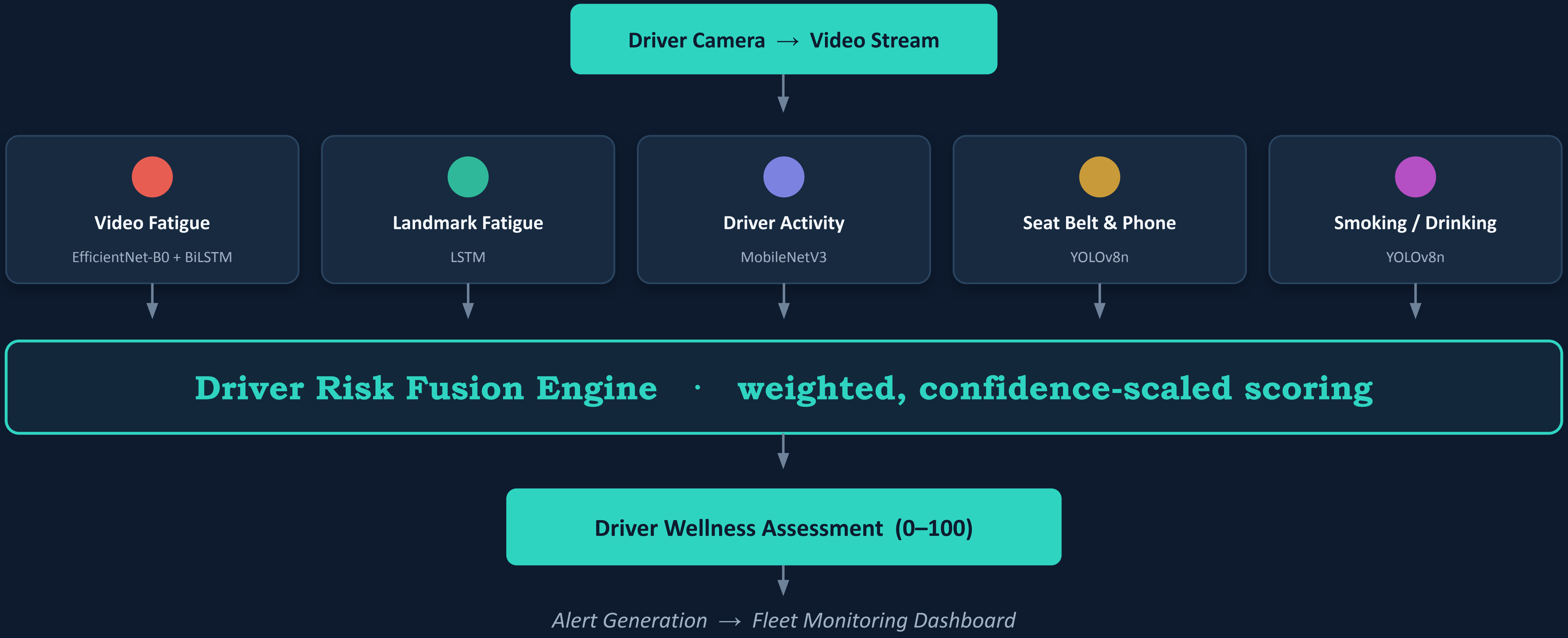
Integrate & Deploy

Standardized adapters, fused 0–100 score, hosted Gradio app.

You are here → Milestone 6 turns five evaluated models into one live, deployed safety signal.

OVERALL SYSTEM

Five Models, One Wellness Score



TEAM & SCOPE

Five Modules, Final Held-Out Results

OWNER		MODULE	MODEL	HELD-OUT HEADLINE
K	**Kushagra**	Video-Based Fatigue	EfficientNet-B0 + BiLSTM	Acc 33.6% · streaming built
S	**Shiwani**	Landmark Fatigue	LSTM (EAR·MAR·pose)	Acc 64.1% · sens. 94.4%
S	**Shubham**	Driver Activity	MobileNetV3-Large	Acc 93.1% · 12.5 ms/frame
S	**Sohini**	Seat Belt & Phone	YOLOv8n	mAP@50 0.95 · P .94 R .90
R	**Ravina**	Smoking & Drinking	YOLOv8n	mAP@50 0.82 · 3.01 M par

All figures are inference-only on strictly held-out, subject- or driver-disjoint splits (Milestone 5).

THE RISK-FUSION ENGINE

Five Signals Fused Into One Live Score

FROM

One model · one label

A discrete, single-cause “fatigue: yes / no” verdict, scored once per clip.

TO

Five models · one score

A fused, continuous 0–100 wellness score, updated every second from the stream.

THE SCORING PIPELINE

1	2	3	4	5	6
Event risk	**Safe = 0**	**Total risk**	**Exponential fuse**	**Smooth**	**Classify**
R_i = severity × confidence	Non-risky labels add nothing	$R\_total = \Sigma R_i$	$100 \times (1 - e^{(-k \cdot R\_total)})$ K=hyperparameter	3-second moving average To reduce fluctuations	Low ⟶ Critical

The engine consumes only each module’s label + confidence — never the internal network — so any model can be swapped or retrained without touching the math.

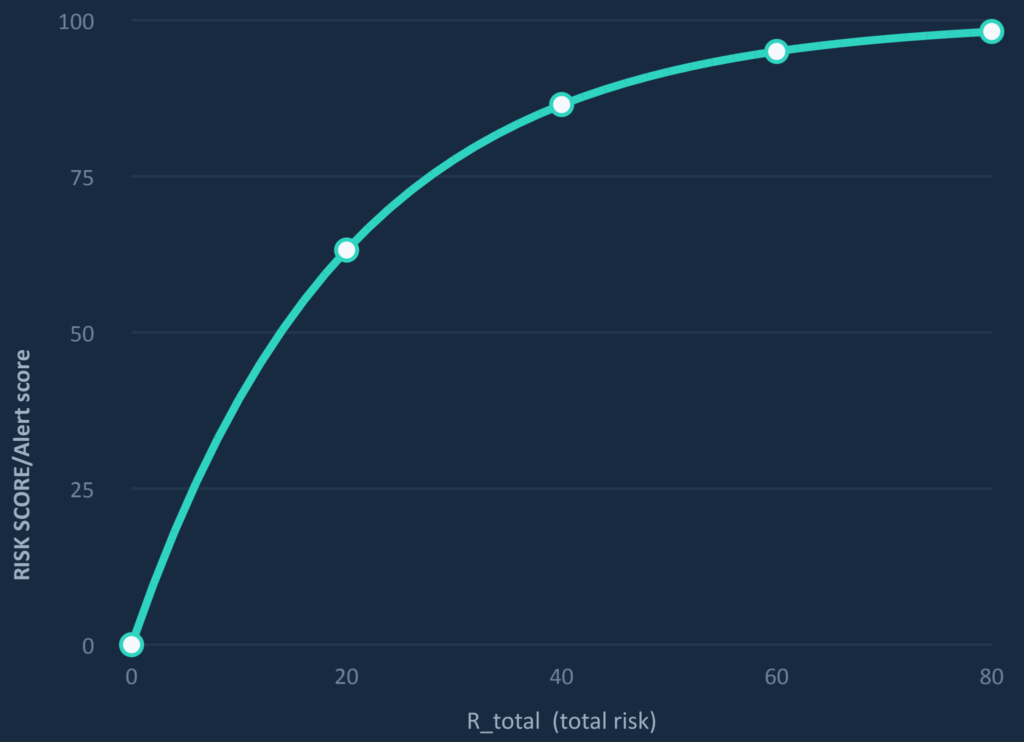
Danger × Certainty, Bounded 0–100

$$R_i = \text{severity\_w}_i \times \text{confidence}_i$$

SEVERITY WEIGHTS (0–10)

Video Fatigue	Medium 6 · High 10
Landmark	Drowsy 7 · Yawning 7 · Talking 4
Driver Activity	Texting 9 · Phone 8 · Other 6 · Turn 4
Seat Belt & Phone	Phone 9 · Phone+Belt 9 · None 8
Smoking / Drinking	Drinking 10 · Both 10 · Smoking 5

Risk score $= 100 \times (1 - e^{-k \cdot R\_total})$ *k = 0.05*



Bounded · additive · saturating — concurrent risks compound, then the curve flattens once danger is already high.

STEPS 5 & 6 · WORKED EXAMPLE

Drowsy, Texting, and Unbelted

MODULE	PREDICTION	SEV	CONF	R_i
Landmark	Drowsy	7	0.90	6.3
Driver Activity	texting_phone	9	0.80	7.2
Seat Belt	No Detection	8	0.75	6.0
Video Fatigue	Low Risk (safe)	0	—	0
Smoking	none (safe)	0	—	0

R_total = 19.5 score = 100 × (1 − e^(−0.05×19.5)) =

62.29 → **HIGH RISK**

SMOOTH → MAP SCORE → RISK LEVEL

Displayed score = mean of raw scores over the last 3.0 s — killing per-frame flicker before the level is chosen.

Low Risk	0 – 25
Moderate Risk	25 – 50
High Risk	50 – 75
Critical Risk	75 – 100

Individual Module → Integrated System

How five independent implementations became one pipeline behind a shared contract.

1. BaseModelAdapter
Common lifecycle / loading contract

2. PredictionResult
Common label + confidence + risk metadata

3. Module Manager
Registers & initializes all five checkpoints

4. Streaming Orchestrator
Decodes frames; applies frame skip / windows

5. Risk Fusion
Consumes standardized outputs only

6. Gradio
Displays predictions, score & session summary

Key engineering idea: each module stays independently maintainable, while the integration layer knows only the standardized contract — not the internal model architecture.

INTEGRATION

End-to-End Integrated Pipeline

One video stream, five adapters, one risk decision.

Driver Video → Streaming Orchestrator → 5 Model Adapters → PredictionResult Objects → Risk Fusion Engine → Wellness Score 0–100

STREAMING BEHAVIOUR

Frame decode → module-specific frame sampling → **temporal windows / smoothing** → standardized result → **risk fusion** → UI update

M6 integration test: 1,760 frames • all five modules active • no module errors • 49.8/100 Moderate Risk

INTEGRATION

End-to-End Validation

Demonstrating that the five modules actually work together.

1,760	**106.7 s**	**5 / 5**	**0**	**49.8 / 100**
Frames processed	Elapsed time	Modules active	Module errors	Observed wellness score

OBSERVED RESULT

MODERATE RISK

Landmark prediction: **Drowsy**

Confidence: **0.831**

IMPORTANT INTERPRETATION

This is an integration / pipeline validation result — not a system-wide accuracy benchmark.

The test confirms that all five modules can contribute to the common risk score, end to end, without failure.

RELIABILITY

Robustness & Failure Handling

The integrated system fails safely and reports module state explicitly.

Unavailable face	Landmark module reports a structured unavailable / error state.
Short video	Insufficient temporal input is handled gracefully.
Model warm-up / failure	Unavailable modules are excluded from fusion rather than treated as safe.
YOLO flicker	Confidence filtering, NMS and temporal stabilization reduce transient detections.
Streaming load	Frame skipping and rolling histories reduce redundant computation.

Gradio Interface

Turning the ML pipeline into a usable driver-monitoring application.

RECORDED VIDEO

- Upload driver video
- Analyze
- Annotated output
- Module predictions
- Wellness gauge
- Session summary

LIVE WEBCAM

- Start webcam
- Streaming inference
- Live annotations
- Real-time score
- Risk-level updates
- Stop + summary

Cockpit-style dashboard • module status chips • risk gauge • visual hierarchy

DEPLOYMENT

Deployment — Lightning AI + Gradio

Packaging the integrated application for hosted inference.

FINAL DEPLOYMENT FLOW

Git / Project → **Lightning AI** → **Gradio App** → **Driver Video** → **Integrated Output**

Git / Project	Lightning AI	Gradio App	Driver Video	Integrated Output
Code + checkpoints	Hosted runtime	User-facing UI	Upload / webcam	Annotations + score

DEPLOYMENT ENGINEERING

- Thin application wrapper + centralized inference core.
- Model caching / lazy loading and frame skipping reduce startup and runtime resource pressure.
- requirements.txt and README.md document the reproducible runtime.
- Gradio exposes the same integrated inference pipeline rather than separate module demos.

Platform note: the consolidated M6 documentation records a hosted Space deployment with runtime quota constraints; this reflects the team’s final Lightning AI hosting update.

Deployment Challenges & Mitigations

What had to be considered beyond model accuracy.

CHALLENGE	IMPACT	MITIGATION / DIRECTION
GPU / runtime quota	Inference may be interrupted or unavailable	Caching, lazy loading, lighter models, dedicated GPU
Five-model memory	Simultaneous checkpoints increase VRAM demand	Lazy initialization; optimize model lifecycle
Long videos	GPU execution time can grow quickly	Frame skipping; shorter windows; future chunking
Landmark throughput	Face-landmark extraction can bottleneck webcam FPS	Optimize MediaPipe / asynchronous processing
Domain shift	Cabin lighting, glare, camera angle affect performance	Broader real-world validation

Deployment readiness is therefore an engineering problem as well as a machine-learning problem.

TEAM

Member Contributions & Ownership

Final contributions across the project lifecycle.

	MEMBER	MODULE	KEY TECHNICAL OWNERSHIP	INTEGRATION CONTRIBUTION
K	Kushagra	Video Fatigue	Sequence modeling; EfficientNet-B0 + BiLSTM; fatigue evaluation	Checkpoint + standardized fatigue output
S	Shiwani	Landmark Fatigue	MediaPipe features; LSTM; temporal fatigue logic	Landmark adapter; temporal state + failure handling
S	Shubham	Driver Activity	MobileNetV3-Large; activity evaluation	DriverActivityAdapter; Gradio lead
R	Ravina	Smoking & Drinking	YOLOv8n; dataset cleaning; tuning; held-out eval	Detection adapter + deployment consolidation
S	Sohini	Seatbelt & Phone	YOLOv8n; robustness; temporal consensus	Adapter; Risk Fusion integration; final docs

Shared responsibility: consistent interfaces, common inference flow, integration testing, documentation and final delivery.

CONCLUSION

Limitations & Where This Goes Next

MODEL LIMITATIONS

- Fatigue generalization across unseen drivers is the largest technical gap.
- Landmark fatigue depends on face visibility and camera orientation.
- Drinking detection is weaker than smoking detection.
- Small-object localization in real cabin conditions remains hard.
- Metrics across modules are task-specific — not directly comparable.

SYSTEM / DEPLOYMENT LIMITATIONS

- Five simultaneous models can require significant GPU memory.
- Live webcam throughput is constrained by landmark extraction.
- Runtime can be constrained by hosted GPU quota / time limits.
- More diverse, longer-duration real-world validation is needed.
- Not safety-certified; a research / engineering prototype.

From drowsiness detection to a complete, fused, deployed driver-wellness system.

Kushagra · Shiwani · Shubham · Sohini · Ravina